

Control of magnetic transitions via interlayer engineering in ferroelectric H₂O–OH systems

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Controlling magnetic phase transitions in two-dimensional (2D) multiferroics is crucial for both fundamental scientific knowledge and practical applications. Here, we present a general strategy for inducing magnetic transitions in 2D ferroelectrics based on the electron pairing principle. First-principles calculations revealed the coupled ferroelectric and ferromagnetic behavior in water-hydroxyl (H₂O–OH) monolayer, with ferromagnetism arising from unpaired electrons in *2p* bonding orbitals of the OH groups. A ferromagnetic-to-nonmagnetic phase transition occurs via interlayer coupling, forming H₂O₂ molecules where electrons pair up. Conversely, the nonmagnetic-to-ferromagnetic transition can be triggered by stacking rearrangements that prevent H₂O₂ formation and restore unpaired electrons. Importantly, such magnetic transitions can be efficiently controlled by external electric fields. Notably, low-energy-electron-assisted synthesis method and high-resolution scanning tunneling microscopy confirm the successful creation of H₂O–OH overlayer on Ag(111). These findings provide an approach for magnetism control in 2D ferroelectrics and offer insights for future spintronic and multiferroic devices.

Two-dimensional (2D) multiferroics, which integrate multiple ferroic orders—such as ferroelectricity and ferromagnetism—within a single material^{1–6}, hold great promise for the development of high-performance, nonvolatile information storage and processing nanodevices. Controlling magnetic phase transitions in 2D multiferroics is essential for achieving higher-density data storage solutions, as it significantly enhances information capacity within a given physical space. Theoretical studies have demonstrated electric field-induced phase transitions from ferromagnetic to antiferromagnetic in ReWCl₆⁷ and NbVS₄⁸ monolayers, both exhibiting strong magnetoelectric coupling. However, magnetic phase transitions in 2D multiferroics remain largely underexplored.

Given the significance of interlayer interactions in layered materials, strategies such as interlayer stacking, sliding, and twisting have become effective in tailoring the properties of materials^{9–21}, including magnetism^{22–32} and ferroelectricity^{33–46}. For example, layer-dependent ferromagnetism has been observed in CrI₃²³, Cr₂Ge₂Te₆²⁴ and CrPS₄²⁵ films at low temperatures. Furthermore, thickness-tunable ferromagnetism has been reported in non-van-der-Waals materials like Cr₅Te₈²⁶ and CrCuSe₂^{27,28}. Theoretical studies have also indicated that altermagnetism can be achieved in 2D antiferromagnetic van der Waals bilayers through twisting^{29,30} and symmetrically restricted direct stacking^{31,32}. Moreover, interlayer engineering has enabled sliding-induced 2D ferroelectricity in the topological semimetal WTe₂⁴³,

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semiconducting 1T'-ReS₂ multilayers⁴⁴, twisted bilayer graphene⁴⁵, and AB-stacking bilayer *h*-BN⁴⁶.

Although layer-dependent magnetism and ferroelectricity in 2D materials have been extensively explored and partially validated experimentally, reports on layer-dependent multiferroicity or sliding-induced magnetic transitions remain scarce. Advancing this field requires a comprehensive investigation into how interlayer interactions influence multiple ferroic orders. A particularly promising class of systems is 2D molecular ferroelectric-ferromagnetic materials, in which intrinsic magnetism arises from unpaired electrons in radical components. In such systems, interlayer engineering can potentially control spin-order transitions via a simple yet fundamental mechanism—electron pairing.

In this work, we realize the layer- and sliding-dependent multiferroicity in the 2D water-hydroxyl (H₂O-OH) systems, identified through first-principles calculations and subsequently synthesized using low-energy-electron-assisted (LEEA) synthesis method. Our calculations reveal coupled ferroelectric and ferromagnetic orders in the freestanding H₂O-OH monolayer: the asymmetric hydrogen-bonding network gives rise to in-plane polarization, while unpaired electrons in the OH group's 2*p* orbitals contribute to ferromagnetism. By strategically stacking two monolayers into a bilayer, interlayer OH-OH interactions lead to the formation of hydrogen peroxide (H₂O₂) molecules, resulting in complete electron pairing and a transition to a nonmagnetic state. Conversely, interlayer sliding prevents H₂O₂ formation, restores unpaired electrons, and induces a transition from nonmagnetism back to ferromagnetism. Furthermore, STM measurements confirm the successful synthesis of H₂O-OH multilayers on the Ag(111) surface.

Results

Ferromagnetism and ferroelectricity in the H₂O-OH monolayer

The freestanding H₂O-OH monolayer adopts an orthorhombic structure with the space group *Amm*2. This structure comprises alternating H₂O molecules and OH groups interconnected through hydrogen bonds, as illustrated in Fig. 1a. Various computational methods, including the HSE06 functional, DFT+U method, meta-GGA with the modified Becke-Johnson potential, and the Perdew-Burke-Ernzerhof hybrid functional (PBE0), were used to calculate the spin-polarized band structures and projected density of states (PDOS). The results indicate that the H₂O-OH monolayer is a ferromagnetic semiconductor (Fig. 1b and Supplementary Fig. 1).

The total magnetic moment for the H₂O-OH monolayer is found to be $-0.79 \mu_B$. A major portion of this moment ($-0.70 \mu_B$) is primarily localized on the O atoms within the OH groups (Fig. 1c), whereas the O atoms in H₂O molecules contribute only a minor amount ($-0.09 \mu_B$). H atoms in the OH groups exhibit an almost negligible magnetic moment of $-0.003 \mu_B$. To further investigate the magnetic properties, three spin configurations were considered on the honeycomb lattice: one ferromagnetic and two antiferromagnetic arrangements. Total energy calculations show that the ferromagnetic configuration is the most stable, with the lowest energy (Supplementary Fig. 2a–c). Furthermore, Monte Carlo simulations estimate a Curie temperature of approximately 11 K for the H₂O-OH monolayer (Supplementary Fig. 2d).

The alternating arrangement of H₂O molecules and OH groups forms an asymmetric hydrogen-bonding network, resulting in the absence of spatial inversion symmetry of the H₂O-OH monolayer. Consequently, a spontaneous electric polarization emerges⁴⁷. The configuration of H₂O and OH groups stabilizes the electric dipoles along in-plane direction, thereby enhancing the polarization effect. Berry phase calculations yield an in-plane polarization of 0.28×10^{-9} C/m along the [110] direction (Supplementary Fig. 3), comparable to other well-studied 2D materials^{6,48–54}, such as SnS (0.26×10^{-9} C/m)⁴⁸, VOI₂ (0.23×10^{-9} C/

m)⁴⁹, Bi₂O₂Se (0.28×10^{-9} C/m)⁵⁰, γ -AlOOH (0.85×10^{-10} C/m)⁵¹ and 1T' CrCoS₄ (0.86×10^{-10} C/m)⁶. This polarization exceeds that of materials exhibiting out-of-plane polarization^{33,46,55–58}.

The kinetics pathways for polarization reversal processes were constructed, and the corresponding energy profiles were estimated to determine the energy barrier for ferroelectric switching in the H₂O-OH monolayer. As a brute-force check, 180° ferroelectric switching involves the displacement of three hydrogen atoms along the directions of their respective hydrogen bonds (highlighted by the black arrows in the insets of Fig. 1d). This leads to an energy barrier between the ferroelectric and intermediate nonpolar states of 0.23 eV for the ferromagnetic state and 0.19 eV for the antiferromagnetic state during the 180° ferroelectric switching (Fig. 1d). During the hydrogen atom shifting process, the intermediate nonpolar state—comprising O[−] anions and hydronium ions (H₃O⁺)—is identified as a metastable phase with an antiferromagnetic ground state (Supplementary Fig. 4). This finding suggests that during ferroelectric switching, a transition from ferromagnetic to antiferromagnetic order may occur, providing a pathway for the magnetic control of intrinsic polarization.

More importantly, the polarization of the H₂O-OH monolayer can be switched by 60°, 120°, 240°, or 300° by directly shifting a single H atom along the hydrogen bond direction (highlighted by the black arrows in the insets of Fig. 1e), which efficiently reduces the energy barrier to 70 meV for the ferromagnetic state (Fig. 1e and Supplementary Fig. 5). It is of particular interest to explore how the magnetization responds to changes in the polarization direction. The magnetic anisotropic energy as a function of the magnetization angle were calculated and presented in Fig. 1f. One can see that the easy axis of magnetization for the H₂O-OH monolayer lies in the plane, with the lowest energy corresponding to in-plane magnetization (Fig. 1f). Interestingly, the easy axis direction can be switched within the *xy* plane as the polarization direction rotates (Fig. 1g–i). This demonstrates the potential for electric-field control of magnetization direction through ferroelectric switching.

Nonmagnetism and ferroelectricity in the H₂O-OH bilayer

The AB-stacked H₂O-OH bilayer represents the global energy minimum, as identified using the CALYPSO method^{59–62}. This structure is formed by shifting the H₂O-OH monolayer along the *c*-axis and rotating it by 60° before assembling it with the original monolayer (Fig. 2a). At this configuration, interlayer OH-OH interactions drive the formation of H₂O₂ molecules (Fig. 2b). The O–O bond length in the resulting H₂O₂ is 1.47 Å, consistent with that in H₂O₂ molecular crystals⁶³. The dynamical stability of the H₂O-OH bilayer has been confirmed by the absence of imaginary frequency modes in the phonon spectra (Supplementary Fig. 6a). Ab initio molecular dynamics (AIMD) simulations support the thermal stability of the H₂O-OH bilayer at room temperature (Supplementary Fig. 6b). Band structures and spin-polarized PDOS calculations (Fig. 2c) reveal that the H₂O-OH bilayer is a nonmagnetic insulator with a wide band gap of 6.58 eV. No spin polarization is observed within the H₂O₂ molecules, in agreement with our calculations, which yield a total magnetic moment of 0 μ_B . These results confirm that the formation of H₂O₂ molecules effectively quenches spin polarization in the bilayer configuration.

The AB-stacked phase belongs to the C₂ space group and exhibits a C₂ point group symmetry. Berry phase calculations indicate that the net in-plane polarization for the H₂O-OH bilayer is identical to that of the monolayer, with a value of 0.28×10^{-9} C/m along the [110] direction (Supplementary Fig. 7). The switching process proceeds via the displacement of four hydrogen atoms—two in the upper layer and two in the lower layer—along the respective directions of the hydrogen bonds, as indicated by the black arrows in the insets of Fig. 2d. Energy

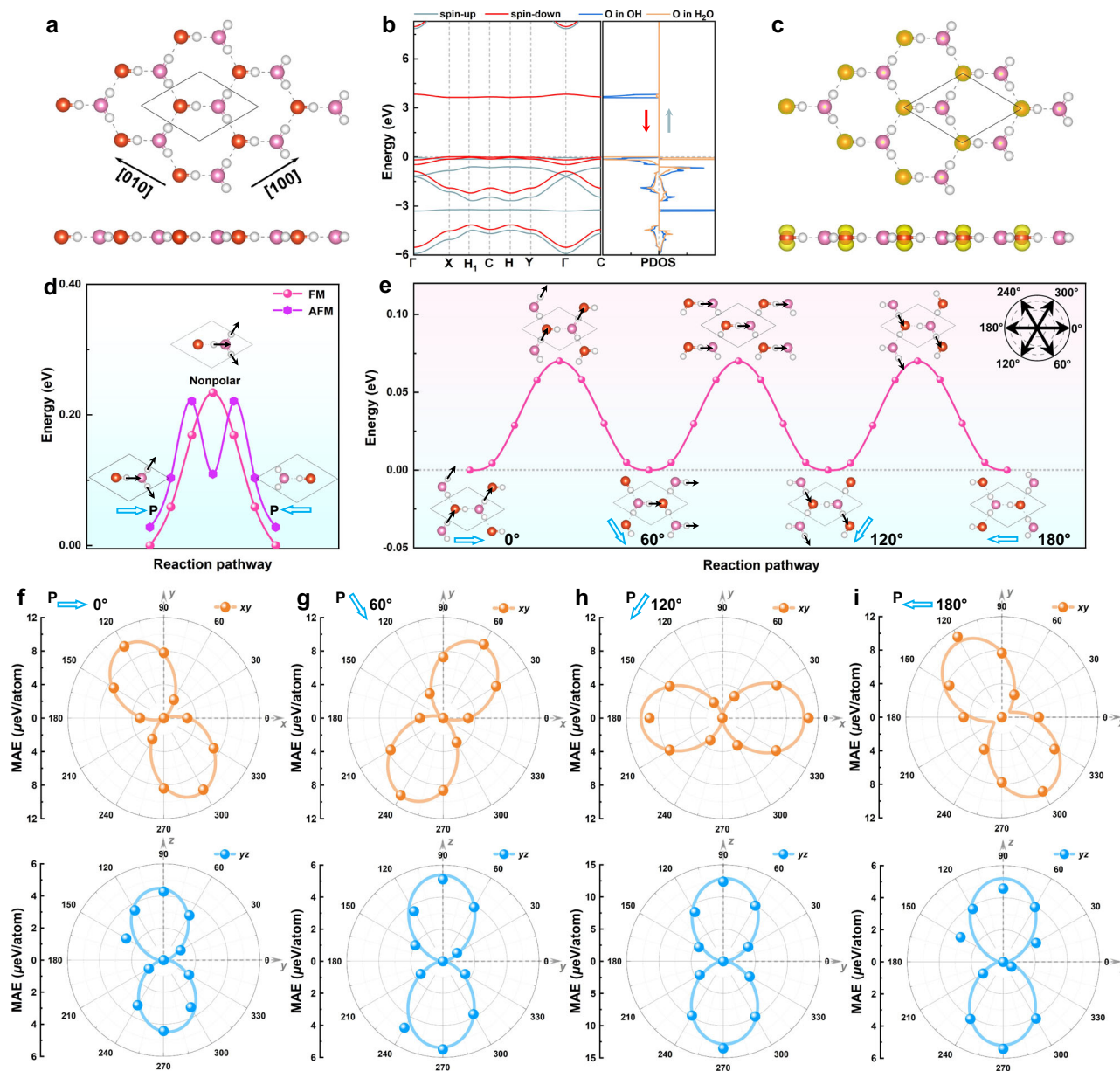


Fig. 1 | Crystal structure, ferromagnetism and ferroelectricity of the H₂O-OH monolayer. **a** Top and side views of the crystal structure. Orange spheres represent O atoms in OH groups, and pink spheres represent O atoms in H₂O molecules. White spheres represent H atoms. **b** Spin-polarized band structure and PDOS calculated using the HSE06 functional. Gray and red curves denote the spin-up and spin-down channels, respectively. **c** Spin density distribution plotted with an iso-surface value of 0.027 e/bohr³. **d, e** Energy profile along the ferroelectric switching pathway, showing the relative energy with respect to the initial

ferroelectric phase. Insets depict the atomic configurations of the initial, transition, and final states during polarization reversal. FM and AFM refer to ferromagnetic and antiferromagnetic states, respectively. Hollow cyan arrows indicate the possible polarization directions, and black arrows indicate atomic displacement during the switching process. **f–i** Angular dependence of the magnetic anisotropic energy (MAE) for polarization orientations of 0°, 60°, 120°, and 180°. Orange and cyan lines indicate the magnetization directions in the xy and yz planes, respectively.

difference calculations between the ferroelectric and intermediate nonpolar states suggest that the energy barrier for 180° ferroelectric switching in the H₂O–OH bilayer is 0.26 eV (Fig. 2d), slightly higher than that of the monolayer.

In addition to 180° switching, the polarization of the bilayer can also be reversed by 60°, 120°, 240°, or 300°, through the direct displacement of two hydrogen atoms along hydrogen bond directions (highlighted in Fig. 2e), reducing the energy barrier to 0.14 eV (Fig. 2e and Supplementary Fig. 8). However, for six-state polarization switching, the energy barrier in the bilayer is approximately twice that of the monolayer due to the involvement of two hydrogen atoms in the process.

Layer- and stacking-dependent magnetic transitions in the ferroelectric H₂O-OH multilayers and the electron pairing mechanism

To better understand the structural and ferroic evolution of H₂O–OH multilayers, we first constructed trilayer and tetralayer configurations with various stack configurations. Our calculations indicate that the ABA-stacked H₂O–OH trilayer possesses the lowest energy (Supplementary Table 1), consisting of an AB-stacked bilayer and an additional monolayer on top (Supplementary Fig. 9a and Fig. 3a). Owing to the weak van der Waals (vdW) interactions between the bilayer and the third layer, the OH groups in the outermost layer retain their magnetic behavior. For the tetralayer H₂O–OH structure, the ABAB-stacked

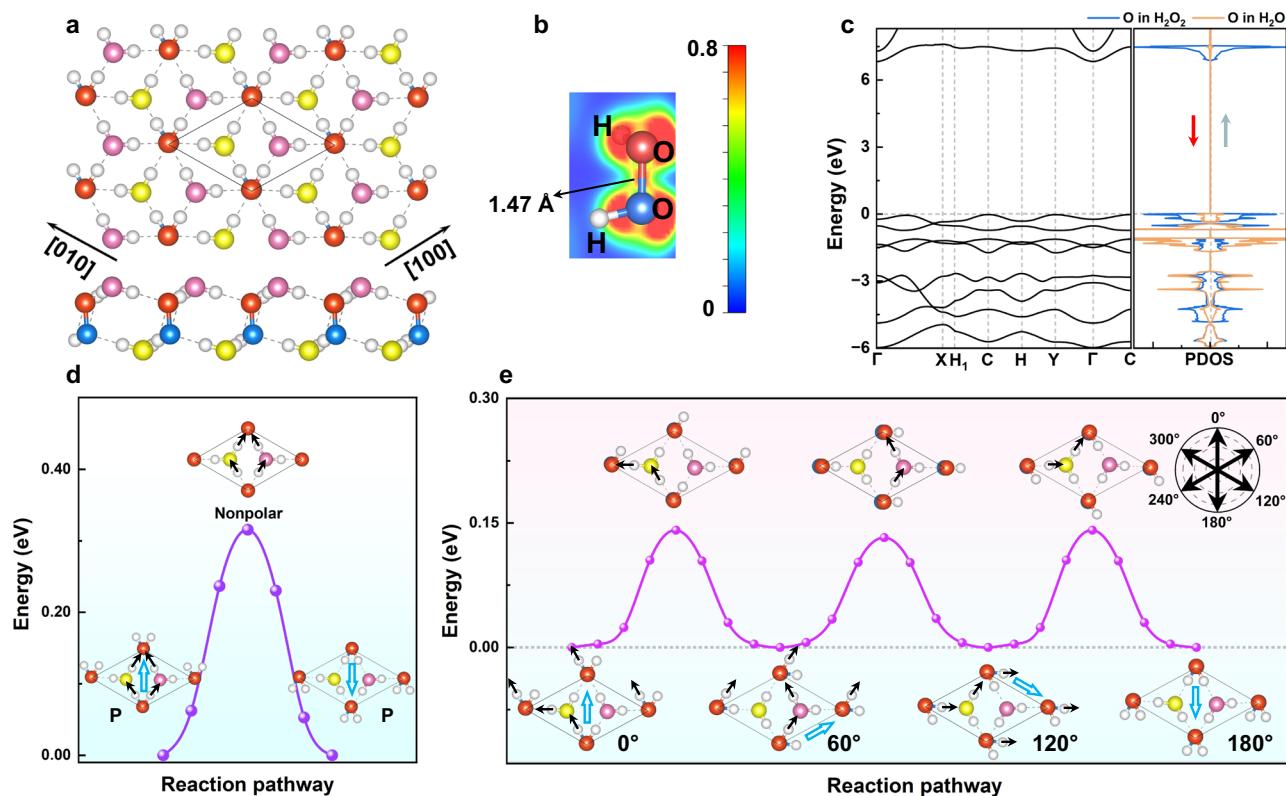


Fig. 2 | Crystal structure, nonmagnetism and ferroelectricity of the H₂O–OH bilayer. **a** Top and side views of the crystal structure. Orange and blue spheres represent O atoms in H₂O₂ molecules; pink and yellow spheres represent O atoms in H₂O molecules; white spheres represent H atoms. **b** 2D electron localization function maps. **c** Band structure and spin-polarized PDOS calculated using the

HSE06 functional. **d, e** Relative energy with respect to the initial ferroelectric phase along the polarization switching pathway of the H₂O–OH bilayer. Insets show atomic configurations of the initial, transition, and final states. Hollow cyan arrows indicate the possible polarization directions, while black arrows attached to atoms depict atomic displacements during the polarization reversal process.

configuration is the most energetically favorable (Supplementary Fig. 9b and Fig. 3a), comprising two AB-stacked bilayers with weak vdW interactions. Phonon spectrum calculations support the dynamical stability of both trilayer and tetralayer structures (Supplementary Fig. 9c, d).

Based on these insights, we further explored multilayer configurations ranging from pentalayer to octalayer (Supplementary Table 2). These structures preserve the asymmetric hydrogen-bonding network, thereby sustaining in-plane polarization across multilayers (Supplementary Fig. 10a–d).

To investigate the evolution of magnetism with increasing layer number, we calculated total energies, structural features, polarization, and magnetic ordering across multilayers, as summarized in Fig. 3a. Spin-polarized electronic structure calculations reveal a striking odd-even oscillation: odd-layer systems exhibit ferromagnetism, whereas even-layer systems are nonmagnetic. The magnetic moments in odd layers mainly originate from O atoms in the OH groups and are comparable to those in the monolayer (Supplementary Fig. 11). Spin density calculations further confirm this trend (Supplementary Fig. 12).

This behavior is well explained by molecular orbital theory. As shown in Fig. 3b, magnetism of the OH groups arises from unpaired electron in the 2p bonding orbitals. Upon formation of H₂O₂ molecules—via interlayer coupling between OH groups—ten electrons (2p⁵ for a pseudoatom of OH) become fully paired, quenching magnetic moments and rendering the structure nonmagnetic. Thus, unpaired electrons in OH groups account for ferromagnetism in odd layers, whereas electron pairing in H₂O₂ molecules leads to nonmagnetism in even layers. This establishes H₂O–OH multilayers as odd-even-dependent multiferroics, emphasizing the central role of electron pairing in determining magnetic order.

Building on this mechanism, we explored the possibility for restoring ferromagnetism in even-layer systems via stacking rearrangement to suppress H₂O₂ formation and retain unpaired electrons. We chose the bilayer as a representative system for detailed study.

As shown in Fig. 3c–f, we constructed four additional stacking configurations—AA, AC, AD, and AE. After full structural relaxation, H₂O₂ formation was found only in the AA-stacked phase, which consequently exhibits nonmagnetism. In contrast, the other three phases, stabilized via vdW interactions, do not form H₂O₂ and thus retain ferromagnetic ground states. The AA-stacked phase is energetically close to the AB-stacked one, with a small energy difference of 6.4 meV/atom, whereas the AC, AD, and AE phases have higher in energy (~200 meV/atom). These energy differences are consistently reproduced across different vdW correction methods (Supplementary Table 3). Phonon spectra validate the dynamical stability of all four stacking configurations (Supplementary Fig. 13). Although the AC-, AD-, and AE-stacked configurations are energetically less favorable, they may still be experimentally accessible via precise stacking control or interlayer sliding techniques.

Symmetry analysis shows that the AA- and AD-stacked bilayers exhibit spontaneous polarization along the [110] direction, while the AB-, AC-, and AE-stacked phases exhibit spontaneous polarization along the [110] direction. A 90° rotation of the electric polarization can be achieved by applying an electric field, which simultaneously induces structural transformations and modifies the magnetic phases. This is confirmed by transitions between the AA- and AC-stacked phases by applying perpendicular electric field (Supplementary Fig. 14a).

Starting from AB- and AA-stacked phases, we slide the upper layer by a distance $d = ma + nb$, where a and b are the lattice vectors, to investigate the sliding-induced transitions. As shown in Fig. 3g, the AC-

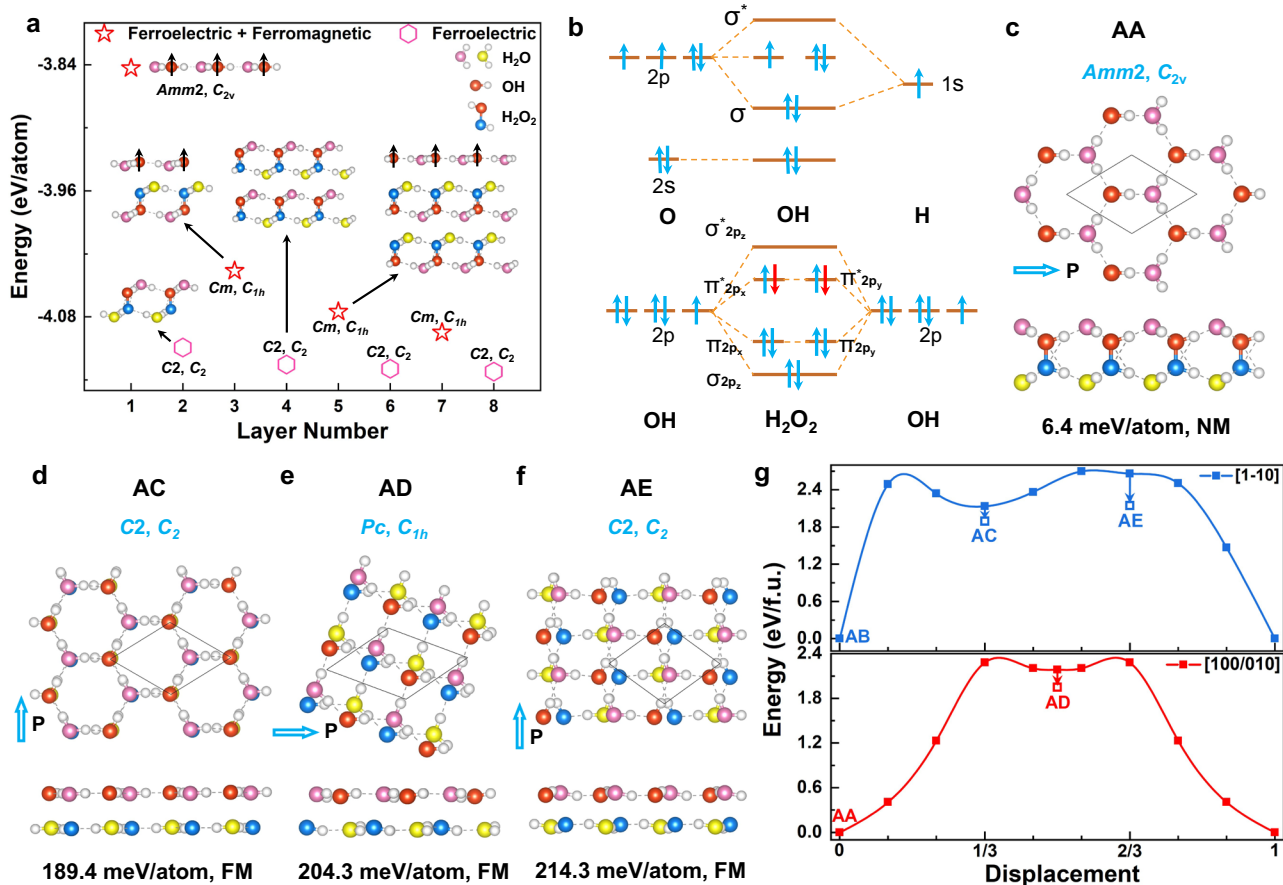


Fig. 3 | Layer- and stacking-dependent magnetic transitions and their mechanism. **a** Total energy of H₂O-OH as a function of layer number from monolayer to octalayer. Insets show the crystal structures of H₂O-OH multilayers, along with their space groups and point groups, and molecular units. **b** The molecular orbital scheme of the OH group and H₂O₂ molecule. The OH group is treated as a pseudoatom with five electrons to construct the molecular orbital diagram of H₂O₂ molecule. **c-f** Crystal structures of the H₂O-OH bilayers with

different stacking configurations. Relative energies (with respect to the AB-stacked phase) and corresponding space and point groups are indicated. FM and NM refer to ferromagnetic and nonmagnetic states, respectively. **g** Sliding energy profiles of the H₂O-OH bilayer with AB and AA stacking along various directions. Interlayer distances were optimized throughout the sliding process. After identifying low-energy configurations, full structural optimization was performed. Hollow squares indicate the energies of the fully optimized H₂O-OH bilayer structures.

and AE-stacked phases can be obtained by shifting the upper layer of the AB-stacked phase by ($m=1/3$, $n=1/3$) and ($m=2/3$, $n=2/3$), respectively. Similarly, the AD-stacked phases emerge from the AA-stacked phase with sliding distances of ($m=1/2$, $n=0$).

Theoretical calculations further predict that an external electric field of -0.43 V/Å along the [110] direction can induce a transition from the AB to AC structure, while a field of -0.25 V/Å along the [110] direction can drive the AA-to-AC transformation (Supplementary Fig. 14b, c). These findings demonstrate that magnetic transitions in the H₂O-OH bilayer can be effectively controlled via interlayer sliding, mediated by externally applied electric fields.

Synthesis of the H₂O-OH multilayers on Ag(111)

Our calculations demonstrated that the $(\sqrt{7} \times \sqrt{7})$ -(H₂O-OH) supercell has a small lattice mismatch of 3.1% with the (4×4) -Ag(111) surface. This minor mismatch reduces the interfacial strain, making the formation of the overlayer energetically favorable. Based on this result, we selected Ag(111) as the substrate for synthesizing the H₂O-OH multilayers using the LEEA method. In this process, low-energy electrons dissociate H₂O at low temperatures, facilitating the formation of H₂O-OH overlayers on the Ag(111) surface. When a small amount of H₂O is deposited on the Ag surface, the combined effects of electron injection and thermal activation result in the formation of a well-ordered H₂O-OH monolayer, as illustrated in Fig. 4a. Increasing the

H₂O coverage leads to the development of multilayer H₂O-OH structures (Supplementary Fig. 15 and Fig. 4b). STM images of these structures on Ag(111) consistently reveal a hexagonal pattern characterized by six bright protrusions. Height profile analyzes confirm that these structures correspond to bilayer, trilayer, and multilayer H₂O-OH formations (Fig. 4b,c).

The lattice constants for the H₂O-OH systems on Ag were determined to be $a = b \approx 11.56$ Å and $\theta \approx 120^\circ$, which corresponds to a 4×4 superstructure of the Ag(111) surface, which is consistent with our structural model for $(\sqrt{7} \times \sqrt{7})$ -(H₂O-OH)/(4 × 4)-Ag(111), as shown in Supplementary Fig. 16a–c. After structural optimization, the calculated step heights closely align with the experimental data (Supplementary Fig. 17). The atomic structures of the topmost layers in the H₂O-OH multilayer exhibit strong similarity to those of the monolayer, resulting in consistent STM images across all formations. The topmost H₂O-OH layer in the trilayer, when superimposed on STM images, aligns precisely with experimental observations (Fig. 4d). Furthermore, the simulated STM image (Supplementary Fig. 18) accurately reproduces the experimental results, displaying the characteristic hexagonal pattern of six bright protrusions and thereby supporting the proposed structural model.

To reveal the electronic structures, we performed scanning tunneling spectroscopy (STS) measurements on monolayer H₂O-OH. As shown in Supplementary Fig. 19, the dI/dV spectrum obtained at the

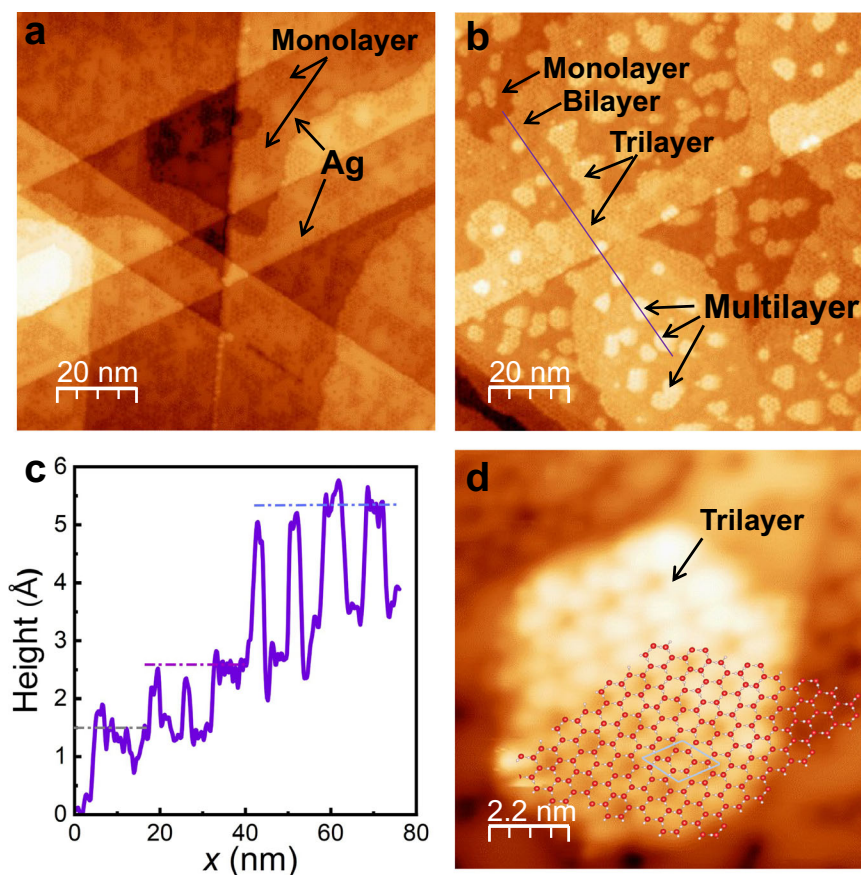


Fig. 4 | STM images showing the evolution of the Ag(111) surface with increasing H₂O coverage exposed to an electron beam of energy 120 eV for 45 s at 70 K, followed by annealing at 275 K. **a STM image of monolayer H₂O-OH on Ag(111) (V = 2 V). **b** STM image showing the coexistence of monolayer, bilayer,**

trilayer, and multilayer H₂O-OH on Ag(111) (V = -0.5 V). **c** Height profiles corresponding to the purple line in (b). **d** A zoomed-in STM image of H₂O-OH trilayer overlaid with a structural model (V = 0.1 V). The blue parallelogram in (d) denotes the ($\sqrt{7} \times \sqrt{7}$) lattice cell.

bright protrusion (marked with a '+' symbol) of the hexagonal ring reveals a characteristic gap of ~ 0.60 eV. The STS shows two shoulders in the DOS, located at 0.4 eV above and 0.95 eV below the Fermi level, corresponding to the conduction band minimum and valence band maximum, respectively. The band edges are notably broadened, making the precise determination of the band gap challenging. Nevertheless, the semiconducting nature of the material is clearly demonstrated.

In summary, we propose a general strategy for achieving transitions between ferromagnetic and nonmagnetic states based on electron pairing principles through interlayer engineering. Using first-principles calculations, we demonstrate that the H₂O-OH monolayer is a 2D multiferroic material, with ferromagnetism arising from unpaired electrons in the bonding orbitals of OH groups. In AB- and AA-stacked bilayer structures, interlayer interactions between OH groups lead to the formation of H₂O₂ molecules, resulting in complete electron pairing and the suppression of magnetism. Consequently, the multiferroicity of H₂O-OH multilayers exhibits an odd-even layer-dependent behavior. By tailoring the stacking configuration to prevent H₂O₂ formation and preserve unpaired electrons, a transition from a non-magnetic to a ferromagnetic state can be induced. Importantly, such magnetic transitions in even-layer bilayers can potentially be controlled by externally applied electric fields, offering a promising route for the realization of multifunctional device applications. Furthermore, we have successfully synthesized H₂O-OH multilayers on the Ag(111) surface, providing experimental validation of the proposed structure. By introducing a route for electric-field-controlled magnetic switching based on the fundamental principle of electron pairing, this work

opens avenues for exploring coupling mechanisms in 2D multiferroics and offers valuable insights for the development of voltage-controlled magnetic switches and non-volatile memory storage devices.

Methods

First-principles calculations

Our first-principles calculations were performed using density functional theory (DFT)⁶⁴ with the projector-augmented wave method (PAW)⁶⁵, as implemented in the Vienna ab initio Simulation Pack (VASP)^{66,67}. The exchange-correlation functional was described by the generalized gradient approximation (GGA) in the form of the Perdew-Burke-Ernzerhof (PBE). The kinetic energy cutoff of the plane wave basis was set to be 650 eV. The first Brillouin zone was sampled using a Γ -centered $7 \times 7 \times 1$ Monkhorst-Pack grid⁶⁸. All structures were relaxed until the convergence criteria for energy and force were 10^{-6} eV and 0.01 eV⁻¹, respectively. We employed several alternative functionals, including DFT+U method, meta-GGA with the modified Becke-Johnson potential, and the Perdew-Burke-Ernzerhof hybrid functional (PBE0), and HSE06 hybrid functional⁶⁹ to estimate the band gap. The optB88-vdW, optB86b-vdW, optPBE-vdW, and vdW-DF methods were used exclusively to describe vdW interactions between layers to identify the optimal stacking order. Electric polarization was calculated using the Berry phase method⁷⁰. A ferroelectric switching pathway was obtained using the climbing-image nudged elastic band method^{71,72}. Phonon spectrum calculations were performed using the finite displacement method, as implemented in the PHONOPY code⁷³. The temperature-dependent effective potential method was used to determine the influence of thermal effects on the phonon dispersion relations⁷⁴. The

external in-plane electric field was applied by imposing a finite homogeneous electric field using the perturbation expression after discretization (PEAD) approach^{75,76}. AIMD simulations were performed using a Nosé–Hoover chain thermostat with a total simulation time of 10 ps (time step of 1 fs).

Sample preparation and STM/STS measurements

Single-crystal Ag(111) was prepared by repeated cycles of Ar ion sputtering and annealing. Ultrapure H₂O (2 ppm; Alfa Aesar) was used and further purified under vacuum by freeze-pump-thaw cycles to eliminate residual impurities. H₂O molecules were dosed in situ onto the pristine Ag(111) surface held at 70 K using a dosing tube. Electron injection was performed over the entire sample surface to ensure homogeneity. The sample was then heated to 275 K, resulting in the appearance of a well-ordered overlayer. Subsequently, the sample was cooled back to 70 K for further measurements. The successful growth of the ordered H₂O–OH monolayer was verified STM. STM measurements were performed in a home-built system at ~5 K. The sample was kept at low temperature during the transfer process. The STS spectra were measured by using lock-in technology with an AC voltage of 20 mV and a superimposed frequency of 667 Hz on the bias voltage. The STM/STS data were processed using the free WSxM software⁷⁷.

Data availability

All data generated in this study are provided in the Supplementary Information.

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Author contributions

M.X., B.F., Y.L. and Y.W. conceived and designed the project and directed the calculations and experiments. J.C. and M.X. performed structure prediction and first-principles calculations. Q.Z., Z.L. and X.H. performed experimental measurements. Y.L., H.J.Z., X.Z., J.F. and L.C. analyzed the data. M.X., B.F. and Y.W. wrote the manuscript with input from all other authors. This manuscript reflects the contributions of all authors.

Competing interests

The authors declare no competing interests. Correspondence should be addressed to M.X., B.F., Y.L., and Y.W.

Additional information

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